

PARALLEL COMPUTING — CLASS 0 DIAGNOSTIC

Prerequisite & problem-solving diagnostic • 2 hours

Purpose: This session checks your current programming, algorithmic, data, and systems knowledge. You are not expected to know parallel computing yet. Work individually and show your reasoning.

Instructions: Use a language you know. Do not worry about perfect syntax unless asked. For algorithm questions, explain your thinking. If you do not know something, say so and move on.

Task 1 — Trace the Code (5 min)

Determine the final output. Show the value of x and y after each statement.

```
x = 5
```

```
y = 2
```

```
x = x + y
```

```
y = x * 2
```

```
x = y - x
```

```
print(x, y)
```

```
1) x = x + y, here 'x' is 7
```

```
2) y = x * 2, here 'y' is 14
```

```
3) x = y - x, here 'x' is 7
```

```
It'll print 7 and 14
```

Task 2 — Find the Bug (5 min)

What does this program actually calculate? What was it probably intended to calculate? Fix it.

This program calculates nothing. It just goes through every number in the “numbers” array by assigning each value to “total” variable each time. It probably intended to calculate the sum of all numbers in the “numbers” array.

```
numbers = [10, 20, 30, 40, 50]
```

```
total = 0
```

```
for i in range(len(numbers)):
    total = numbers[i]
```

```
print(total)
```

To fix that we can just add '+' sign before '=' sign by making incrementation

So the fixed part is 4th line -> `total += numbers[i]`

Task 3 — Nested Loops (5 min)

For the program below: (a) how many times does `print()` execute? (b) what if both loops run from 0 to 999? (c)

what if the outer loop runs 10 times and the inner loop runs 1,000,000 times?

a) 100 times

b) $1000 * 1000$ times, which is 1, 000, 000 times

c) $10 * 1, 000, 000$ times, which is 10, 000, 000 times

```
for i in range(10):
    for j in range(10):
        print(i, j)
```

Part B — Algorithms & Complexity

Task 4 — Which Is Faster? (5 min)

Algorithm A scans an array once. Algorithm B compares every element with every element. For an array of size n , estimate the amount of work for each. Which grows faster? What happens when $n = 1,000,000$?

Algorithm A's estimated amount of the work is $O(n)$, 1, 000, 000. Because it scans each element once.

Algorithm B's estimated amount of the work is $O(n^2)$ which is two times 1, 000, 000.

So 'B' grows faster. And by comparing with 'A' it is very big difference.

Task 5 — Complexity Ranking (7 min)

Rank these from fastest-growing to slowest-growing as n

increases: n^2 $\log_2 n$ n $n \log_2 n$ 1 2^n

Which become particularly problematic when n becomes much larger?

1; n ; $\log_2 n$; $n \log_2 n$; n^2 ; 2^n .

2^n becomes particularly problematic when n becomes much larger

Task 6 — Searching (7 min)

Given [3, 8, 12, 17, 24, 31, 45, 51, 63], find 45 using two approaches: (1) if you do not know the list is sorted; (2) if you know it is sorted. Explain why the second can be faster.

(1) In that case we can find out the target number by just comparing with each element of the array.

(2) If we know that the array is sorted, we can find it faster by comparing every time the middle element, and by that we remove a lot of values that are bigger or smaller than the target number.

Part C — Data & Problem Solving

Task 7 — Large Dataset (8 min)

You have 500 million integers and need to find the largest number. Design an algorithm. How many times does each number need to be examined? Do you need to store all 500 million numbers? Could you process the data

in chunks?

The algorithm is to group these integers by group making them chunks. By that way

Task 8 — Counting (5 min)

Given A B A C B A D C A B, design an algorithm producing $A \rightarrow 4$, $B \rightarrow 3$, $C \rightarrow 2$, $D \rightarrow 1$. What data structure would you use?

I would use map, HashMap<Character, Integer>

Task 9 — Matrix Operations (5 min)

$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$. Calculate $A+B$ and $A \times B$. How many individual multiplication operations are required for two 2×2 matrices?

$A+B = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}$

$A \times B = \begin{bmatrix} 12 & 22 \\ 43 & 50 \end{bmatrix}$

Part D — Systems Intuition

Task 10 — CPU vs Memory (6 min)

A program scans a huge array, but the CPU is extremely fast and the program is still slow. Give three possible reasons.

-
1. Lack of memory
 2. Huge sized array
 - 3.
-

```
for x in huge_array:
    total += x
```

Task 11 — What Happens When You Run a Program? (7 min)

Describe what happens between typing `./program` and seeing the program's output. Use ordinary language. Include anything you think is relevant.

-
1. Compiles program
 2. Run the program code
 3. Program's output
-

Task 12 — Four Programs (7 min)

Four programs run simultaneously: A uses lots of CPU; B uses lots of memory; C reads a large file; D mostly waits for user input. Identify the likely bottleneck for each. Can all four run at the same time? What determines CPU time? What changes if there is one CPU core?

-
- A. There might be problem with other parts, where also needed CPU
 - B. Same as the 'A', so there might be a lack of memory which cause slow
 - C. It might be very big time, CPU, memory consuming process
 - D. Here is the problem when user input is very big
-

We can start run them at the same time, but it will slow down our computer and cause instability.

If there is one CPU core we can't run program simultaneously

Part E — Reasoning Challenges

Task 13 — The Slow Program (6 min)

A program processes 1 billion records in 10 hours. A student says: “If we buy a computer that is twice as fast, the program will take 5 hours.” Do you agree? Explain.

Nope. There is the logic that at the moment we reach the nearly boarded speed limit, after that limit the boost helps will not work.

Task 14 — Make It Faster (6 min)

A program takes 60 seconds. You may change anything except the answer it produces. List as many ways as possible to reduce execution time.

1. Checking loop usages
2. Memory usage
3. Algorithms that used

Task 15 — Bottleneck Identification (6 min)

Execution times are: A=2s, B=2s, C=50s, D=3s, E=2s. Total=59s. If A, B, D and E become 10× faster, how much faster does the entire program become? Where should optimization effort go?

50 seconds and 0,9 seconds which is 9 / 10.

Final Challenge — Design the Algorithm

Task 16 — 10 Million Student Marks (10 min)

You receive 10 million student marks, each between 0 and 100. You must produce: minimum, maximum, average, median, number scoring ≥ 90 , and number scoring < 40 . Design an algorithm; identify data structures; estimate how many times the data must be examined; estimate memory requirements; identify the likely bottleneck; and explain how you would make it faster without changing the answer.

Use a 101-size frequency array to process all 10 million marks in a single pass using only 404 bytes of memory, shifting the bottleneck entirely to I/O speed which can be accelerated via parallel chunk-processing

STOP HERE. Check that you attempted every task. You are not expected to know parallel computing yet.